Word-Level Estimation of Shannon and Rényi Entropy Improves Reading Time Predictions

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Entropy as a Predictor of Sentence Processing Difficulty

- Contextual entropy (Eqns. 1, 3) is often used as a measure of the difficulty of predicting the next word during language comprehension [1, 7, 10, 4]
- Contrasts with surprisal, which measures difficulty of the *current* word
- Language models (LMs) like GPT-2 [8] can be used to estimate entropy
- Because calculating exact word-level entropy from an LM is intractable, entropy is typically computed over the first subword token instead
- We instead estimate word-level entropy using a sampling technique

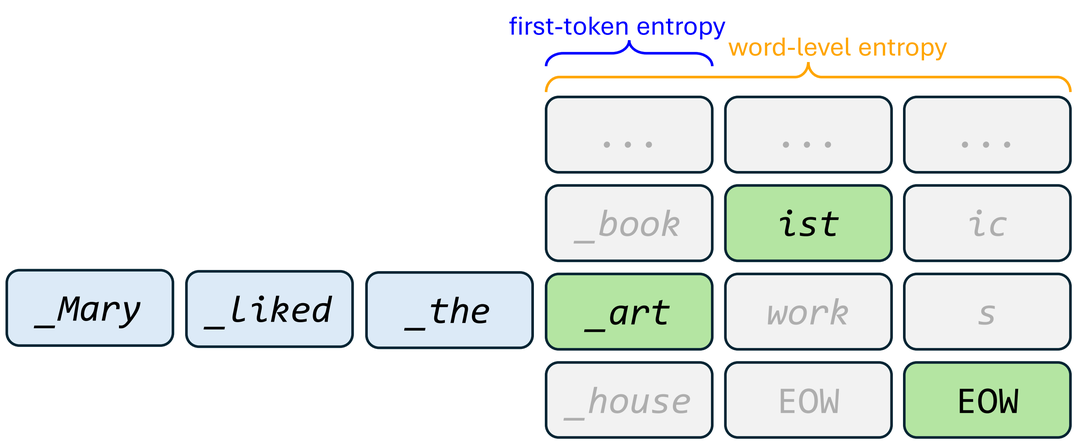

Figure 1: Illustration of first-token and word-level entropy estimation. The latter involves sampling multi-token words such as *_artist* in this example.

- Word-level entropy is higher than first-token entropy, especially for open classes like nouns (NN, NNS) and adjectives (JJ)

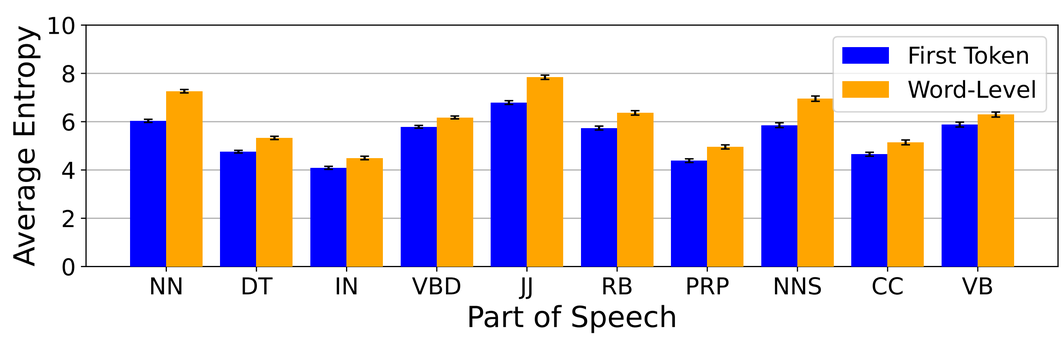

Figure 2: Average Shannon entropy of the 10 most common parts of speech in the Natural Stories corpus [3], using either first-token or word-level entropy approximations.

Shannon Entropy Estimation

- Definition of contextual Shannon entropy:

$$\mathrm{H}(W_i \mid w_{1..i-1}) = -\sum_{w \in V} \mathrm{P}(w \mid w_{1..i-1}) \log_2 \mathrm{P}(w \mid w_{1..i-1}) \quad (1)$$

  - W_i is current word, $w_{1..i}$ is context, V is vocabulary
  - Typically estimated over a word's first subword token
- Sampling-based estimate of contextual Shannon entropy:

$$\mathrm{H}(W_i \mid w_{1..i-1}) \approx -\frac{1}{|S|} \sum_{s \in S} \log_2 \mathrm{P}(s \mid w_{1..i-1}) \quad (2)$$

  - S is a multiset of sampled next words (Fig. 1)

Rényi Entropy Estimation

- Rényi entropy is a generalization of Shannon entropy capturing a range of possible anticipatory processing strategies [7]
- Definition of contextual Rényi entropy of order α:

$$\mathrm{H}_\alpha(W_i \mid w_{1..i-1}) = \lim_{\beta \to \alpha} \frac{1}{1-\beta} \log_2 \sum_{w \in V} \Big(\mathrm{P}(w \mid w_{1..i-1}) \Big)^{\beta} \quad (3)$$

  - $\alpha = 0$: Rényi entropy equals number of possible next words
  - $\alpha = 1$: Rényi entropy equals Shannon entropy (by L'Hôpital's rule)
  - $\alpha = \infty$: Rényi entropy equals surprisal of most likely next word
- Sampling-based estimate of contextual Rényi entropy:

$$\mathrm{H}_\alpha(W_i \mid w_{1..i-1}) \approx \lim_{\beta \to \alpha} \frac{1}{1-\beta} \log_2 \left(\frac{1}{|S|} \sum_{s \in S} \Big(\mathrm{P}(s \mid w_{1..i-1}) \Big)^{\beta - 1} \right) \quad (4)$$

Regression Experiments

- Predicted reading times from five psycholinguistic corpora
  - Self-paced reading (SPR): Natural Stories, Brown [3, 9]
  - Eye tracking: Dundee, Provo, GECO [5, 6, 2]
  - Eye-tracking corpora included first-pass (FP) and go-past (GP) durations
- Fit mixed-effects models with and without word entropy predictors
- $\alpha = 1/2$ was used for Rényi entropy, following Pimentel et al. [7]
- Baseline predictors in regression models: word length, word index, unigram surprisal, LM surprisal, fixation on previous word
- Entropy and surprisal predictors from GPT-2 [8]
- Evaluation metric: Per-datapoint log likelihood increase (ΔLL)
  - Compares likelihood of regression models with and without an entropy predictor
  - Calculated for both first-token and word-level entropy estimates
  - Averaged over k-fold cross-validation with $k = 10$

Results

- Aggregated results (Fig. 3) show significant improvement in ΔLL from replacing first-token Shannon and Rényi entropy with word-level entropy
- Mixed results across individual corpora for Shannon entropy (Table 1a)
- Numerically consistent improvements from word-level estimation for Rényi entropy (Table 1b)

- Stronger predictions from Rényi entropy suggest that humans use a different anticipatory processing strategy from Shannon entropy
- Differences between token- and word-level predictors show a need for caution in relying on token-level approximations

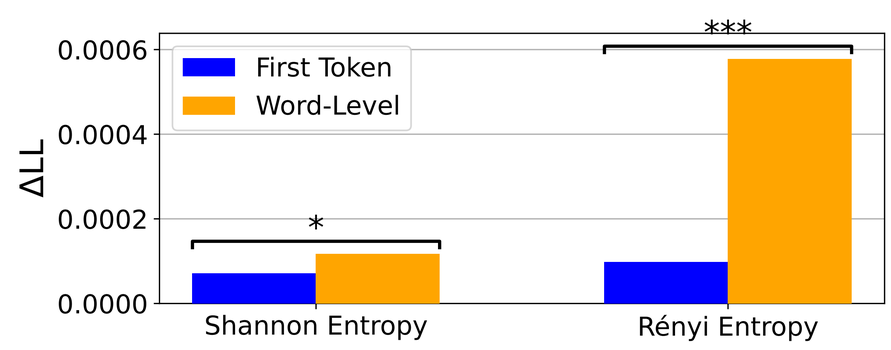

Figure 3: Improvements in reading time predictions from first-token and word-level estimates of Shannon and Rényi entropy. These results are aggregated over all reading time corpora.

Corpus	$\Delta\mathrm{LL}_{\mathrm{FT}}$	$\Delta\mathrm{LL}_{\mathrm{WL}}$
NS SPR	1.26×10^{-4}	3.05×10^{-4} ***
Brown SPR	2.10×10^{-6}	-1.75×10^{-5}
Dundee FP	1.28×10^{-5}	-4.09×10^{-6}
Dundee GP	9.21×10^{-6}	-4.60×10^{-6}
Provo FP	-7.54×10^{-6}	1.18×10^{-5}
Provo GP	2.13×10^{-4} **	6.48×10^{-5}
GECO FP	8.11×10^{-5} ***	1.17×10^{-5}
GECO GP	-1.38×10^{-6}	-3.45×10^{-6}
Combined	7.09×10^{-5}	1.17×10^{-4} *

(a) Shannon Entropy

Corpus	$\Delta\mathrm{LL}_{\mathrm{FT}}$	$\Delta\mathrm{LL}_{\mathrm{WL}}$
NS SPR	2.61×10^{-4}	1.43×10^{-3} ***
Brown SPR	-1.05×10^{-5}	1.99×10^{-4}
Dundee FP	1.02×10^{-6}	1.10×10^{-4} **
Dundee GP	-4.09×10^{-6}	1.90×10^{-4} **
Provo FP	-1.29×10^{-5}	-8.95×10^{-6}
Provo GP	4.51×10^{-5}	6.57×10^{-5}
GECO FP	5.86×10^{-6}	2.76×10^{-5}
GECO GP	0.00	1.14×10^{-5}
Combined	9.85×10^{-5}	5.78×10^{-4} ***

(b) Rényi Entropy

Table 1: Per-corpus improvements in reading time predictions. $\Delta\mathrm{LL}_{\mathrm{FT}}$ and $\Delta\mathrm{LL}_{\mathrm{WL}}$ respectively refer to log likelihood improvements from first-token and word-level entropy. NS means Natural Stories.

Acknowledgments and References

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